Un nouveau moyen d'approcher la conjecture d'Erdos Straus

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Abstract

The Erdős-Straus conjecture states that for every integer $n \geq 2$, the equation $\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ admits positive integer solutions. While historical approaches, notably by Mordell, have relied on quadratic residues and polynomial identities, this paper proposes a novel and independent method based on a complete congruence network.

We analyze the equation, specifically for $n \equiv 1 \pmod 4$, by classifying primes through congruences modulo 3 and 7. We establish a series of theorems that provide explicit algorithms to construct solutions for these specific classes. Furthermore, we demonstrate that for any prime n, the conjecture is equivalent to the existence of an integer $c \equiv 3 \pmod 4$ such that c divides the sum of two divisors of $K = \frac{n+c}{4}$. This result offers a new arithmetic reduction of the problem.

Finally, to validate these theoretical findings, we developed a high-performance computational verification algorithm in C, optimizing the solution search up to $n = 10^{10}$.

Keywords: Erdős-Straus conjecture, Egyptian fractions, Diophantine equations, Modular arithmetic, Number theory, Computational Verification.

1 Introduction

The Erdős-Straus conjecture, formulated by Paul Erdős and Ernst G. Straus in 1948, proposes that for every integer $n \geq 2$, there exist positive integers x, y and z such that:

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

This conjecture is closely related to other problems in number theory, notably those involving Egyptian fractions, such as the Sierpiński conjecture. As early as 1950, the mathematician Richard Obláth took up the calculations of Straus and Harold N. Shapiro and verified that the conjecture holds true for $n \leq 10^5$. Subsequently, other mathematicians worked on the problem, such as Luigi Rosati (1954), Ernest Kiss (1959), Léon Bernstein (1962), Koichi Yamamoto (1965), Ralph Jollensten (1976), and Swett in 1999, who showed that the conjecture holds for $n \leq 10^{14}$. But the most important work of the 20th century on this conjecture surely remains that of Louis Mordell, who proved in 1967 that the conjecture is true for $n \not\equiv 1, 11^2, 13^2, 17^2, 19^2, 23^2 \pmod{840}$.

In 2009, Michel Mizony proved that the conjecture is true from an algorithmic point of view. In 2011, Terence Tao and Christian Elsholtz published a new idea, that of classifying the solutions into two types (type 1 and type 2). These works would later inspire Miguel Angel Lopez, who published the latest research on the conjecture in 2022. In it, he develops the idea of a complete congruence network associated with the solutions, which he distinguishes into two types (type A and type B).

Since Mordell's work, the use of quadratic residues and polynomials has become the preferred method for approaching this problem. I decided to

start with a different, more accessible approach. Since this approach has nothing to do with previous advances, we will reprove everything from the beginning and in an independent manner. The idea behind this work is to successfully find a solution to the given equation for $n \equiv 1 \pmod 4$ by going through the congruence (mod 7).

Théorème 1 *: For every non-zero even natural integer n, we have:*

$$\frac{4}{n} = \frac{1}{n} + \frac{1}{n} + \frac{1}{\frac{n}{2}}$$

Proof : Let n be a non-zero even natural integer, there exists $h \in \mathbb{N}^*$ such that $n = 2h$

$$\frac{1}{n} + \frac{1}{n} + \frac{1}{\frac{n}{2}} = \frac{1}{2h} + \frac{1}{2h} + \frac{1}{h} = \frac{4}{2h} = \frac{4}{n}$$

Théorème 2 *: For every non-zero odd natural integer n, there exists a $k \in \mathbb{N} \setminus \{0, 1\}$ such that:*

$$\frac{1}{k} < \frac{4}{n} < \frac{1}{k-1}$$

Proof : Let n be a non-zero odd natural integer. According to the Archimedean property, there exists $k \in \mathbb{N} \setminus \{0, 1\}$ such that $4(k-1) < n < 4k$ By strict decrease of $4k \mapsto \frac{1}{4k}$ on $\mathbb{R}^*$:

$$\frac{1}{4(k-1)} > \frac{1}{n} > \frac{1}{k} \Leftrightarrow \frac{1}{k-1} > \frac{4}{n} > \frac{1}{k}$$

Lemme 1 *: From the previous proof we have $n < 4k$, so there necessarily exists an integer j such that $n + j = 4k$*

Théorème 3 *: For all $n \equiv 3 \pmod 4$ we have:*

$$\frac{4}{n} = \frac{1}{k} + \frac{1}{2kn} + \frac{1}{2kn}$$

Proof : Let $n \equiv 3 \pmod 4$:

$$\frac{1}{k} + \frac{1}{2kn} + \frac{1}{2kn} = \frac{1}{k} + \frac{1}{kn} = \frac{n+1}{kn}$$

Now $n + j = 4k \Leftrightarrow k = \frac{n+j}{4}$

Thus
$$\frac{n+1}{kn} = \frac{n+1}{\frac{n(n+j)}{4}} = \frac{4(n+1)}{n(n+j)}$$

Now $n \equiv 3 \pmod 4$ therefore $j = 1$

Thus
$$\frac{4(n+1)}{n(n+j)} = \frac{4}{n}$$

2 $n \equiv 1 \pmod 4$

Théorème 4 *: For all* $n \equiv 1 \pmod 4$ *:*

$$\frac{4}{n} = \frac{1}{kn} + \frac{2}{kn} + \frac{1}{k}$$

Proof : Let $n \equiv 1 \pmod 4$

$$\frac{1}{kn} + \frac{2}{kn} + \frac{1}{k} = \frac{3}{kn} + \frac{1}{k} = \frac{3+n}{kn} = \frac{3+n}{\frac{n(n+j)}{4}} = \frac{4(3+n)}{n(n+j)}$$

Now $n \equiv 1 \pmod 4$ therefore $j = 3$ Thus

$$\frac{4(3+n)}{n(n+j)} = \frac{4}{n}$$

This formula cannot strictly speaking be used in resolving this conjecture because the presence of the 2 in the numerator of the sum: $\frac{1}{kn} + \frac{2}{kn} + \frac{1}{k}$ is problematic. We can, however, use it to solve more complex cases.

Corollaire 1 *: For all* $n \equiv 1 \pmod 4$ *and* $kn \equiv 0 \pmod 2$ *:*

$$\frac{4}{n} = \frac{1}{kn} + \frac{1}{\frac{kn}{2}} + \frac{1}{k}$$

Corollaire 2 *: For all* $n \equiv 1 \pmod 4$ *and* $kn \equiv 0 \pmod 3$ *:*

$$\frac{4}{n} = \frac{1}{\frac{2kn}{3}} + \frac{1}{\frac{2kn}{3}} + \frac{1}{k}$$

Proof :

$$\frac{4}{n} = \frac{1}{kn} + \frac{2}{kn} + \frac{1}{k} = \frac{2}{2kn} + \frac{4}{2kn} + \frac{1}{k} = \frac{3}{2kn} + \frac{3}{2kn} + \frac{1}{k}$$

Thus

$$\frac{4}{n} = \frac{1}{\frac{2kn}{3}} + \frac{1}{\frac{2kn}{3}} + \frac{1}{k}$$

Corollaire 3 *: For all* $n \equiv 1 \pmod 4$ *and* $kn \equiv 0 \pmod 5$ *:*

$$\frac{4}{n} = \frac{1}{\frac{2kn}{5}} + \frac{1}{2kn} + \frac{1}{k}$$

Proof :

$$\frac{4}{n} = \frac{1}{kn} + \frac{2}{kn} + \frac{1}{k} = \frac{2}{2kn} + \frac{4}{2kn} + \frac{1}{k} = \frac{5}{2kn} + \frac{1}{2kn} + \frac{1}{k}$$

Thus

$$\frac{4}{n} = \frac{1}{\frac{2kn}{5}} + \frac{1}{2kn} + \frac{1}{k}$$

The problem posed by this conjecture essentially relies on the fact that $kn \equiv 0 \pmod p$ where p is a prime number different from $2, 3$ and 5. We will subsequently seek to find a solution to this problem by classifying the prime numbers through a congruence network.

Théorème 5 *: For every* $n \equiv 1 \pmod 4$ *and* ($k \equiv 2 \pmod 3$ *or* $n \equiv 2$

(mod 3)), *there exists a non-zero natural integer o such that:*

$$\frac{4}{n} = \frac{1}{on} + \frac{1}{okn} + \frac{1}{k}$$

Proof : Let $n \equiv 1 \pmod 4$, such that ($k \equiv 2 \pmod 3$ or $n \equiv 2 \pmod 3$), let us set $o = \frac{k+1}{3}$. According to *Theorem 4*:

$$\frac{4}{n} = \frac{1}{kn} + \frac{2}{kn} + \frac{1}{k} = \frac{o}{okn} + \frac{2o}{okn} + \frac{1}{k} = \frac{3o}{okn} + \frac{1}{k} = \frac{k+1}{okn} + \frac{1}{k} = \frac{k}{okn} + \frac{1}{okn} + \frac{1}{k}$$

This proof could also have been conducted if $n \equiv 2 \pmod 3$, by setting $o = \frac{n+1}{3}$.

Théorème 6 *: For every $n \equiv 1 \pmod 4$, if $n + k \equiv 0 \pmod 3$ then there exists a non-zero natural integer o' such that:*

$$\frac{4}{n} = \frac{1}{o'k} + \frac{1}{o'n} + \frac{1}{k}$$

Proof : Let $n \equiv 1 \pmod 4$, such that $n + k \equiv 0 \pmod 3$. Let us set $o' = \frac{k+n}{3}$, according to *Theorem 4*:

$$\frac{4}{n} = \frac{1}{kn} + \frac{2}{kn} + \frac{1}{k} = \frac{o'}{o'kn} + \frac{2o'}{o'kn} + \frac{1}{k} = \frac{3o'}{o'kn} + \frac{1}{k} = \frac{n+k}{o'kn} + \frac{1}{k} = \frac{n}{o'kn} + \frac{k}{o'kn} + \frac{1}{k} = \frac{1}{o'k} + \frac{1}{o'n} + \frac{1}{k}$$

This theorem is in reality just another way of expressing Theorem 5.

We will now proceed with an example where $kn \equiv 0 \pmod p$, and where it is impossible for us to use one of the previous theorems. This will allow us to intuitively understand the upcoming concepts.

Example 1 : Let us set $n = 73$
We have $72 < 73 < 76 \Leftrightarrow \frac{1}{72} > \frac{1}{73} > \frac{1}{76} \Leftrightarrow \frac{1}{18} > \frac{4}{73} > \frac{1}{19}$ $73 \equiv 1 \pmod 4$,

thus according to *Theorem 4*:

$$\frac{4}{73} = \frac{1}{19} + \frac{2}{1387} + \frac{1}{1387}$$

$73 \not\equiv 2 \pmod 3$ and $19 \not\equiv 2 \pmod 3$, it is impossible to use *Theorem 5*.

$n+k = 73+19 = 92$ but $92 \not\equiv 0 \pmod 3$, it is impossible to use *Theorem 6*.

Let us try to modify our formula with $k+1$ instead of k, that is to say 20 instead of 19 in our example. We then notice that:

$$\frac{4}{73} = \frac{1}{20}+\frac{1}{73\times 6}+\frac{1}{73\times 6}+\frac{1}{73\times 10\times 6} = \frac{1}{20}+\frac{1}{73\times 6}+\frac{1}{73\times 6}+\frac{5}{73\times 10\times 6}-\frac{4}{73\times 10\times 6}$$

$$= \frac{1}{20}+\left(\frac{1}{73\times 6}+\frac{5}{73\times 10\times 6}\right)+\left(\frac{1}{73\times 6}-\frac{4}{73\times 10\times 6}\right) = \frac{1}{20}+\frac{1}{292}+\frac{1}{730}$$

We deduce from this example the method to follow when we cannot use theorems *5 and 6*. The underlying idea is that we need to change k and use $k+1$, if that still doesn't work we will take $k+2$ and so on.

3 $n \equiv 1 \pmod 4$, $n \equiv u \pmod 7$, $kn \equiv 0 \pmod p$

The objective now is to treat by case disjunction the set of u belonging to $\mathbb{N}$ such that $u \in \{0, 1, \ldots, 6\}$ and where $n \equiv 1 \pmod 4$, $n \equiv 1 \pmod 3$, $n \equiv u \pmod 7$, $k \equiv 1 \pmod 3$ and $kn \equiv 0 \pmod p$ where p is a prime number different from $2, 3$ and 5 (since these cases have already been covered).

Theorem 7 : *For every* $n \equiv 1 \pmod 3$, $n \equiv 1 \pmod 4$, $n \equiv 3 \pmod 7$ *and* $k \equiv 1 \pmod 3$ *and* $kn \equiv 0 \pmod p$, *there exists* $\varepsilon \in \mathbb{N}^*$, *such that:*

$$\frac{4}{n} = \frac{1}{\frac{n\varepsilon}{2}} + \frac{1}{k+1} + \frac{1}{\frac{(k+1)\varepsilon n}{2}}$$

Proof : Let $n \equiv 1 \pmod 3$, $n \equiv 1 \pmod 4$, $n \equiv 3 \pmod 7$ and $k \equiv 1 \pmod 3$ and $kn \equiv 0 \pmod p$.

Let us set $\varepsilon = \frac{n+11}{14}$, let's verify that $\varepsilon \in \mathbb{N}^*$.

n is odd so $n+11$ is divisible by 2, $11 \equiv 4 \pmod 7$ and $n \equiv 3 \pmod 7$ so $n + 11 \equiv 0 \pmod 7$. Thus $\varepsilon \in \mathbb{N}^*$.

$$\varepsilon = \frac{n+11}{14} \iff \varepsilon = \frac{-n-11}{-14} \iff -7 = \frac{-n-11}{2\varepsilon} \iff -7 = \frac{-n-7}{2\varepsilon} - \frac{2}{\varepsilon}$$

$$\iff -7 + n = \frac{-n-7}{2\varepsilon} - \frac{2}{\varepsilon} + n \iff n = \frac{-n-7}{2\varepsilon} - \frac{2}{\varepsilon} + n + 7$$

Now $\frac{n+3}{4} = k$ therefore $\frac{n+7}{4} = k+1$

Thus

$$n = -\frac{2(k+1)}{\varepsilon} - \frac{2}{\varepsilon} + 4(k+1) \iff n + \frac{2(k+1)}{\varepsilon} + \frac{2}{\varepsilon} = 4(k+1) \iff n\varepsilon + 2(k+1) + 2 = 4(k+1)\varepsilon$$

Now

$$\frac{n\varepsilon + 2(k+1) + 2}{n\varepsilon(k+1)} = \frac{1}{\frac{n\varepsilon}{2}} + \frac{1}{k+1} + \frac{1}{\frac{(k+1)\varepsilon n}{2}}$$

Therefore

$$\frac{4(k+1)\varepsilon}{n\varepsilon(k+1)} = \frac{1}{\frac{n\varepsilon}{2}} + \frac{1}{k+1} + \frac{1}{\frac{(k+1)\varepsilon n}{2}}$$

Thus

$$\frac{4}{n} = \frac{1}{\frac{n\varepsilon}{2}} + \frac{1}{k+1} + \frac{1}{\frac{(k+1)\varepsilon n}{2}}$$

We can ensure that $\frac{(k+1)\varepsilon n}{2} \in \mathbb{N}^*$ because $kn \equiv 1 \pmod 2$ therefore $k \equiv 1 \pmod 2$ therefore $k+1 \equiv 0 \pmod 2$.

We can ensure that $\frac{n\varepsilon}{2} \in \mathbb{N}^*$, because $\varepsilon = \frac{n+11}{14}$ and that $n \equiv 1 \pmod 4$ and $11 \equiv 3 \pmod 4$, thus $\frac{n+11}{2} \equiv 0 \pmod 2$, and since $\varepsilon \in \mathbb{N}^*$, $\frac{n+11}{14} \equiv 0 \pmod 2$

Example 2 : *Let us set* $n = 241$

$$240 < 241 < 244 \Leftrightarrow \frac{1}{240} > \frac{1}{241} > \frac{1}{244} \Leftrightarrow \frac{1}{60} > \frac{4}{241} > \frac{1}{61}$$

$\varepsilon = \frac{241+11}{14} = 18$

$$\frac{4}{n} = \frac{1}{(k+1)} + \frac{1}{n\epsilon} + \frac{1}{\frac{(k+1)n\epsilon}{4}}$$

We have $(k+1) = 62$

Thus

$$\frac{4}{241} = \frac{1}{194} + \frac{1}{21532} + \frac{1}{2088604}$$

Theorem 8 : *For every* $n \equiv 1 \pmod 3$, $n \equiv 1 \pmod 4$ *and* $n \equiv 5 \pmod 7$ *and* $k \equiv 1 \pmod 3$ *and* $kn \equiv 0 \pmod p$, *there exists* $\lambda \in \mathbb{N}^*$, *such that:*

$$\frac{4}{n} = \frac{1}{k+1} + \frac{1}{n\lambda} + \frac{1}{\frac{(k+1)n\lambda}{4}}$$

Proof : Let $n \equiv 1 \pmod 3$, $n \equiv 1 \pmod 4$ and $n \equiv 5 \pmod 7$ and $k \equiv 1 \pmod 3$ and $kn \equiv 0 \pmod p$.

Let us set $\lambda = \frac{n+23}{28}$, let's verify that $\lambda \in \mathbb{N}^*$.

We have: $n \equiv 1 \pmod 4$ and $n \equiv 5 \pmod 7$.

Applying the Chinese remainder theorem: $M = 4 \times 7 = 28$ $M_1 = 7$ and $M_2 = 4$ $y_1 = 3$ because $7 \times 3 \equiv 1 \pmod 4$ and $y_2 = 2$ because $4 \times 2 \equiv 1 \pmod 7$

$$n \equiv a_1 M_1 y_1 + a_2 M_2 y_2 \pmod M$$

$$n \equiv 1 \times 7 \times 3 + 5 \times 4 \times 2 \pmod{28}$$

Therefore $n \equiv 61 \pmod{28}$ therefore $n \equiv 5 \pmod{28}$ Thus $n + 23 \equiv 0 \pmod{28}$, $\lambda \in \mathbb{N}^*$.

$$\lambda = \frac{n+23}{28} \iff 7 = \frac{n+23}{4\lambda} \iff n = 7+n-(\frac{n+23}{4\lambda}) \iff n = 7+n-(\frac{n+7}{4\lambda}+\frac{4}{\lambda})$$

$$\iff n = 4(k+1) - \frac{k+1}{\lambda} - \frac{4}{\lambda} \iff n = \frac{4(k+1)\lambda - (k+1) - 4}{\lambda}$$

$$\iff n\lambda = 4(k+1)\lambda - (k+1) - 4 \iff n\lambda + (k+1) + 4 = 4(k+1)\lambda$$

Now

$$\frac{n\lambda + (k+1) + 4}{n\lambda(k+1)} = \frac{1}{k+1} + \frac{1}{n\lambda} + \frac{1}{\frac{(k+1)n\lambda}{4}}$$

Therefore

$$\frac{4(k+1)\lambda}{n\lambda(k+1)} = \frac{1}{k+1} + \frac{1}{n\lambda} + \frac{1}{\frac{(k+1)n\lambda}{4}}$$

Thus

$$\frac{4}{n} = \frac{1}{k+1} + \frac{1}{n\lambda} + \frac{1}{\frac{(k+1)n\lambda}{4}}$$

We can ensure that $\frac{(k+1)n\lambda}{4}$ belongs to $\mathbb{N}^*$, $k+1 \equiv 0 \pmod 2$, therefore

$$\frac{(k+1)n\lambda}{4} \in \mathbb{N}^* \iff \frac{n\lambda}{2} \in \mathbb{N}^* \text{ but } kn \equiv 1 \pmod 2 \text{ therefore } n \equiv 1 \pmod 2$$

Thus

$$\frac{n\lambda}{2} \in \mathbb{N}^* \iff \frac{\lambda}{2} \in \mathbb{N}^*$$

Let us show that $n \equiv 1 \pmod 8$.

Assume that $n \equiv 5 \pmod 8$, $n+3 \equiv 0 \pmod 8$ therefore $n+3 = 8w$ (with $w \in \mathbb{N}^*$)

Now

$$\frac{n+3}{4} = k \text{ therefore } k = 2w, \text{ but } kn \equiv 1 \pmod 2, \text{ impossible. Thus } n \equiv 1 \pmod 8.$$

$$23 \equiv 7 \pmod 8 \text{ therefore } 23 + n \equiv 0 \pmod 8$$

Therefore

$$\frac{23+n}{28} \equiv 0 \pmod 2.$$

Thus

$$\frac{\lambda}{2} \in \mathbb{N}^*.$$

Theorem 9 : *For all* $n \equiv 1 \pmod 3$, $n \equiv 1 \pmod 4$ *and* $n \equiv 6 \pmod 7$, k *is prime, there exists* $(k+1)$ *and* ϕ *belonging to* $\mathbb{N}^*$, *such that :*

$$\frac{4}{n} = \frac{1}{k+1} + \frac{1}{n\phi} + \frac{1}{\frac{(k+1)n\phi}{4}}$$

We will deliberately not prove this theorem as the proof is similar to the previous one, which adds no major interest. Let us use an example to illustrate

this formula instead.

$$\phi = \frac{15+n}{28}$$

Example 3 : *Let us set* $n = 769$

$$768 < 769 < 772 \Leftrightarrow \frac{1}{192} > \frac{4}{n} > \frac{1}{193}$$

$$(k+1) = k+1 = 194$$

$$\phi = \frac{769+15}{28} = 28$$

$$\frac{4}{n} = \frac{1}{k+1} + \frac{1}{n\phi} + \frac{1}{\frac{(k+1)n\phi}{4}}$$

$$\text{Thus} \quad \frac{4}{769} = \frac{1}{194} + \frac{1}{21532} + \frac{1}{2088604}.$$

Lemma 2 : *For all* k *belonging to* $\mathbb{N}^*$*, let us set* K *and* α *belonging to* $\mathbb{N}^*$ *such that* $K = k + \alpha$*.*

For all K *belonging to* $\mathbb{N}^*$*, there exists* c *belonging to* $\mathbb{N}^*$ *such that :*

$$K = \frac{n+c}{4}.$$

Proof : Let K, k and α belonging to $\mathbb{N}^*$. Let us set $c = j + 4\alpha$

$$K = k + \alpha = \frac{n+j}{4} + \alpha = \frac{n+j+4\alpha}{4} = \frac{n+c}{4}$$

It is towards the end of this research work that I began to take an interest in the work of other mathematicians on the subject, by reading the beginning of the work: *Sur la conjecture d'Erdös et Straus* by Michel Mizony (2009) that I realize I missed an extremely important and very simple lemma :

Lemma 3 : *If n satisfies the conjecture then $v \times n$ does as well, because if :*

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \text{ then : } \frac{4}{v \times n} = \frac{1}{v \times x} + \frac{1}{v \times y} + \frac{1}{v \times z}$$

Thus any multiple of a prime number n satisfying the conjecture also satisfies it.

We will therefore take n to be prime from now on.

Theorem 10 : *For all $n \equiv 1 \pmod 3$, $n \equiv 1 \pmod 4$ and $n \equiv 2 \pmod 7$, $kn \equiv 0 \pmod p$, $k \equiv 1 \pmod 3$, there exists K and σ belonging to $\mathbb{N}^*$, such that :*

$$\frac{4}{n} = \frac{1}{K} + \frac{1}{\sigma n} + \frac{1}{\frac{n\sigma K}{c\sigma - K}}$$

This theorem also concerns $n \equiv 4 \pmod 7$ and $n \equiv 1 \pmod 7$, and the idea behind the proof is the same so these cases will not be studied.

Proof approach : Let $n \equiv 1 \pmod 3$, $n \equiv 1 \pmod 4$ and $n \equiv 2 \pmod 7$, $k \equiv 1 \pmod 3$, $kn \equiv 0 \pmod p$, K and σ belonging to $\mathbb{N}^*$

$$\frac{1}{K} + \frac{1}{\sigma n} + \frac{1}{\frac{n\sigma K}{c\sigma - K}} = \frac{1}{K} + \frac{1}{\sigma n} + \frac{c\sigma - K}{n\sigma K} = \frac{\sigma n + K + c\sigma - K}{n\sigma K} = \frac{n + c}{nK}$$

$$\text{Now } n + c = 4K, \text{ thus } \frac{n + c}{nK} = \frac{4}{n}$$

Therefore $\frac{4}{n} = \frac{1}{K} + \frac{1}{\sigma n} + \frac{1}{\frac{n\sigma K}{c\sigma - K}}$ is true for all K and σ belonging to $\mathbb{N}^*$ satisfying $\frac{n\sigma K}{c\sigma - K} \in \mathbb{N}^*$

We can pause for a moment on this equation :

$$\frac{n + c}{nK} = \frac{4}{n}$$

This means that

$$\frac{4}{n} = \frac{1}{K} + \frac{c}{nK} \iff \frac{4}{n} = \frac{1}{K} + \frac{4c}{n(n+c)}$$

It would therefore be necessary to prove that $\frac{4c}{n(n+c)}$ can be decomposed into a sum of two unit fractions.

Let us fix n as prime, suppose that $c\sigma - K \mid n\sigma K$, since n is prime we then have $c\sigma - K \mid \sigma K$, so there exists a non-zero natural integer t, such that :

$$tc\sigma - tK = \sigma K \iff tc\sigma = K(t+\sigma) \iff \frac{tc\sigma}{K} = t+\sigma \iff \frac{tc\sigma}{K} - t = \sigma \iff 1 = \frac{tc}{K} - \frac{t}{\sigma}$$

$$\iff \frac{tc}{K} - 1 = \frac{t}{\sigma} \iff \sigma = \frac{t}{\frac{tc}{K} - 1} \iff \sigma = \frac{tK}{tc - K} \iff \sigma = \frac{K}{c - \frac{K}{t}}$$

By definition $t = \frac{K\sigma}{c\sigma - K}$ so t is not coprime to K (in other words t shares a common divisor strictly greater than 1 with K) unless $c\sigma - K = K$, in which case $\sigma = \frac{2K}{c}$.

According to the fundamental theorem of arithmetic, $K = p_1 \times p_2 \times \ldots \times p_m$.

Thus $t = a \times p_i$ where p_i is a divisor of K (which can be a prime factor or a product of prime factors) and a is a non-zero natural integer that does not divide K (because otherwise a would be included in p_i) or else it is in excess with respect to p_i.

Therefore $\frac{K}{t}$ is merely a product of prime factors of K divided by a. We then set $\frac{K}{t} = \frac{d_2}{a}$.

$$\sigma = \frac{K}{c - \frac{K}{t}} \iff \sigma = \frac{K}{c - \frac{d_2}{a}} \iff \sigma = \frac{Ka}{ca - d_2}$$

Assume for the sake of contradiction that $ca - d_2 \mid a$, with $ca - d_2$ different

from 1, then there exists $l \in \mathbb{N}^*$ such that :

$$a = l(ca - d_2) \iff a = lca - ld_2 \iff a(lc-1) = ld_2 \iff \frac{ld_2}{a} = lc - 1$$

Now $a = l(ca - d_2) \iff \frac{l}{a} = \frac{1}{ca-d_2}$ so a does not divide l.

And $\frac{ld_2}{a} = lc - 1$, therefore $a \mid d_2$, according to Gauss's lemma, but by definition a does not divide d_2, which is impossible.

Thus, if $\frac{Ka}{ca-d_2} \in \mathbb{N}$, it means that $ca - d_2 \mid K$.

Therefore $ca - d_2$ is a divisor of K and we set $d_1 = ca - d_2$.

$$\begin{aligned} d_1 + d_2 = ca &\iff a = \frac{d_1 + d_2}{c} \\ &\iff Ka = \frac{K(d_1 + d_2)}{c} \\ &\iff \frac{Ka}{d_1} = \frac{K}{d_1}\left(\frac{d_1 + d_2}{c}\right) \\ &\iff \sigma = \frac{K}{d_1}\left(\frac{d_1 + d_2}{c}\right) \end{aligned}$$

Let us show that d_1 is different from d_2 : Suppose that $d_1 = d_2 \iff ca - d_2 = d_2 \iff ca = 2d_2 \iff a = \frac{2d_2}{c}$.

c is coprime to 2 so according to Gauss's lemma, $c \mid d_2$ therefore $\frac{n+c}{4c} \in \mathbb{N}^*$.

Therefore $\frac{1}{4}\left(\frac{n}{c} + 1\right) \in \mathbb{N}^*$ which is absurd since n is prime. We therefore conclude that if K is prime, it is impossible for $c\sigma - K \mid \sigma K$.

The heart of the problem of the Erdős-Straus conjecture can then be summarized by the following proposition: there exists $\sigma \in \mathbb{N}^*$ if and only if there exist two divisors of K, d_2 and d_1, whose sum is a multiple of c.

Example 4 :

$n = 20353$, 20353 is prime, $20353 \equiv 1 \pmod 4$,

$20353 \equiv 1 \pmod 3$, $20353 \equiv 4 \pmod 7$.

Let us set $c = 23$, $K = \frac{20353+23}{4} = 5094$.

$5094 = 2 \times 3 \times 283$

2 is a divisor of 5094 and $849 = 3 \times 283$ is also a divisor of 5094.

$849 + 2 = 851$

and $851 = 23 \times 37$

c divides the sum of two divisors of K.

$$\sigma = \frac{K}{t} \times \frac{t + d_2}{c} \quad \text{therefore} \quad \sigma = \frac{5094}{849} \times \frac{849 + 2}{23} = 222$$

$$\frac{4}{n} = \frac{1}{K} + \frac{1}{\sigma n} + \frac{1}{\frac{n\sigma K}{c\sigma - K}}$$

Thus

$$\frac{4}{20353} = \frac{1}{5094} + \frac{1}{222 \times 20353} + \frac{1}{\frac{222 \times 20353 \times 5094}{23 \times 222 - 5094}}$$

Another approach would be to set $\sigma = \frac{n+4c-bc-r}{4c}$, where r is the remainder of the Euclidean division of n by c, such that $0 \leq r < c$ and $b = 0$ if $r \equiv 1 \mod 4$, $b = 1$ if $r \equiv 2 \mod 4$, and $b = 2$ if $r \equiv 3 \mod 4$. If $r \equiv 0 \mod 4$, we will assume that no σ exists for such a c and we will try with the next one.

$n = qc + r$ therefore $n - r = qc$, thus $\sigma = \frac{q-b+4}{4}$ and therefore $q \equiv b \mod 4$.

Showing that $c\sigma - K \mid \sigma K$ then amounts to showing that:

$$c\left(\frac{n + 4c - bc - r}{4c}\right) - \frac{n + c}{4} \mid \left(\frac{4 - b + q}{4}\right) \frac{n + c}{4} \iff 3c - bc - r \mid (4 - b + q)\, \frac{n + c}{4}.$$

Assuming this to be true, we then have the equality:

$$\frac{4 - b + q}{4} = \frac{Ka}{ca - d_2} \iff 4 - b + q = \frac{(n + c)a}{ca - d_2} \iff \frac{n + c}{4 - b + q} = \frac{ca - d_2}{a}$$

Now by definition $\dfrac{d_2}{a} = \dfrac{K}{t}$ therefore $\dfrac{n+c}{4-b+q} = c - \dfrac{K}{t} \iff \dfrac{n+c}{4t} = c - \dfrac{n+c}{4-b+q}$

$$\iff \frac{4t}{n+c} = \frac{c\,(4-b+q) - (n+c)}{(n+c)\,(4-b+q)}.$$

Thus,

$$\frac{1}{4t} = \frac{c\,(4-b+q) - (n+c)}{(n+c)\,(4-b+q)} \iff \frac{1}{4t} = \frac{c}{n+c} - \frac{1}{4-b+q} \iff \frac{1}{4t} + \frac{1}{4-b+q} = \frac{c}{n+c}.$$

Since $4-b+q$ is always divisible by 4, we have just given a decomposition of $\frac{4c}{n+c}$ into two unit fractions, we then have:

$$\frac{4}{n} = \frac{1}{K} + \frac{c}{nK} \iff \frac{4}{n} = \frac{1}{K} + \frac{4c}{n(n+c)} \iff \frac{4}{n} = \frac{1}{K} + \frac{1}{nt} + \frac{1}{n\sigma}$$

where t is a non-zero natural integer. But this result was expected because $t = \frac{\sigma K}{c\sigma - K}$.

4 An error in the conjecture!?

I created a program that allowed me to verify that $c\sigma - K$ divides σK and therefore that the conjecture holds true for all tested n up to 10^{10}. I considered $\sigma = \frac{q-b+4}{4}$ to achieve this. With the exception of $n = 3361$, for which there is no value of c where $c\sigma - K$ divides K. But this does not mean that the Erdős-Straus conjecture is false. Indeed, there are several possible solutions for this number, and we are going to see one:

By taking $c = 3$, we study $n \equiv 1 \pmod 3$. Thus we know that $r = 1$ so $b = 0$ and we have $k = \frac{3361+3}{4} = 841$.

$$\sigma = \frac{3361 + 4 \times 841 - 1}{4 \times 3} = 281$$

We have $c\sigma - k = \frac{3\times 281-841}{4} = 2$, but $c\sigma - k$ cannot divide $k\sigma$ because here $k\sigma \equiv 1 \pmod 2$. We could move on to the next c, but the problem would remain the same (the next c admitting a solution is 39 for anyone who wants to try).

However, if we had taken $\sigma = 290$, we would then have $c\sigma - k = 29$. It is then obvious that $c\sigma - k$ divides σk.

The problem stems directly from the formula $\sigma = \frac{n+4c-bc-r}{4c}$. In reality, in certain cases, b can also take negative values provided it keeps the same congruence $\pmod 4$. In our case, we could then have taken $b = -36$, which gives

$$\sigma = \frac{3361 + 40 \times 3 - 1}{4 \times 3} = 290.$$

But the values of b are very difficult to predict, which is why I had restricted myself strictly to the values of $b : \{0, 1, 2\}$.
Another way to solve the problem would have been to use *Theorem 4:*

$$\frac{4}{3361} = \frac{1}{841} + \frac{2}{841 \times 3361} + \frac{1}{841 \times 3361} \iff \frac{4}{3361} = \frac{1}{841} + \frac{3}{841 \times 3361}.$$

By noting that $841 = 29^2$, we could then have considered doing:

$$\frac{4}{3361} = \frac{1}{841} + \frac{3 \times 10}{841 \times 3361 \times 10} \iff \frac{1}{841} + \frac{1}{29 \times 3361 \times 10} + \frac{1}{841 \times 3361 \times 10}.$$

This result was expected, since 1 and 29 are divisors of k, 30 = 29 + 1 and 30 is a multiple of c = 3 so we necessarily had $c\sigma - k$ dividing σk.

5 Unification of methods and strict determination of b

This difficulty in fixing b *a priori* shows the limits of a purely empirical approach. To solve this problem and lift this indeterminacy, I decided to unify our methods by combining the fractional approach and the Euclidean

approach. As we demonstrated previously, the expression for σ is:

$$\sigma = \frac{K}{d_1}\left(\frac{d_1 + d_2}{c}\right)$$

On the other hand, the Euclidean algorithmic approach (based on the division $n = qc + r$) gave us another expression for the same σ:

$$\sigma = \frac{n + 4c - bc - r}{4c}$$

Theorem 11 : *For any $K = \frac{n+c}{4}$, by unifying the fractional approach and the Euclidean approach, the parameter b satisfies the exact identity:*

$$b = 3 - \frac{r + 4K\frac{d_2}{d_1}}{c}$$

Proof : Since these two formulas calculate the same quantity σ, we can equate them:

$$\frac{K}{d_1}\left(\frac{d_1 + d_2}{c}\right) = \frac{n + 4c - bc - r}{4c}$$

We will now inject our master equation $n = 4K - c$ into the numerator on the right:

$$\frac{K}{d_1}\left(\frac{d_1 + d_2}{c}\right) = \frac{(4K - c) + 4c - bc - r}{4c} \iff \frac{K}{d_1}\left(\frac{d_1 + d_2}{c}\right) = \frac{4K + 3c - bc - r}{4c}$$

We will expand the left term by separating the fraction $\frac{d_1+d_2}{d_1}$ to reveal the ratio of the divisors:

$$\frac{K}{c}\left(1 + \frac{d_2}{d_1}\right) = \frac{4K + 3c - bc - r}{4c}$$

To eliminate the denominators, we multiply everything by $4c$:

$$4K\left(1 + \frac{d_2}{d_1}\right) = 4K + 3c - bc - r \iff 4K + 4K\frac{d_2}{d_1} = 4K + 3c - bc - r$$

We notice that the term $4K$ is present on both sides. We subtract it, which simplifies the equation:

$$4K\frac{d_2}{d_1} = 3c - bc - r$$

All that remains is to isolate the term bc by moving it to the left, and sending the rest to the right:

$$bc = 3c - r - 4K\frac{d_2}{d_1}$$

Thus, by dividing everything by c, we obtain our final formula:

$$b = 3 - \frac{r + 4K\frac{d_2}{d_1}}{c}$$

Following new algorithmic tests, we posit as a verified empirical postulate that the parameter $b \in \{0, 1, 2\}$.

Theorem 12 : *The transfer identity of the parameter b simplifies into a strict linear Diophantine equation:*

$$d_1(3 - b) - d_2(q + 1) = a \cdot r$$

Proof : Let us resume our exact identity for the parameter b:

$$b = 3 - \frac{r + 4K\frac{d_2}{d_1}}{c} \iff 3 - b = \frac{r + 4K\frac{d_2}{d_1}}{c} \iff c(3 - b) = r + 4K\frac{d_2}{d_1}$$

Let us multiply everything by d_1 to clear the fraction:

$$c \cdot d_1(3 - b) = r \cdot d_1 + 4K \cdot d_2$$

Now the Euclidean division gives us $n = qc + r$. By definition $4K = n + c$, therefore $4K = qc + r + c = c(q + 1) + r$. We inject this expression for $4K$:

$$c \cdot d_1(3-b) = r \cdot d_1 + (c(q+1)+r)d_2 \iff c \cdot d_1(3-b) = r \cdot d_1 + c(q+1)d_2 + r \cdot d_2$$

Let us move the terms with c to one side and the terms with r to the other:

$$c \cdot d_1(3-b) - c(q+1)d_2 = r \cdot d_1 + r \cdot d_2 \iff c[d_1(3-b) - d_2(q+1)] = r(d_1 + d_2)$$

Now, our fundamental condition stipulates that $a \cdot c = d_1 + d_2$. We therefore replace the sum of the divisors:

$$c[d_1(3-b) - d_2(q+1)] = r(a \cdot c)$$

Since c is a non-zero natural integer, we can divide by c on both sides:

$$d_1(3-b) - d_2(q+1) = a \cdot r$$

Lemma 4 : *From the previous proof, since $a \geq 1$ and $r > 0$, we necessarily have $a \cdot r > 0$.*

Thus $d_1(3-b) - d_2(q+1) > 0 \iff d_1(3-b) > d_2(q+1) \iff \frac{d_2}{d_1} < \frac{3-b}{q+1}$. Since $q \approx n/c$ grows massively towards infinity, this inequality demonstrates a radical and mandatory asymmetry: $d_2 \ll d_1$.

We can pause for a moment on this equation:

$$d_1 = \frac{a \cdot r + d_2(q+1)}{3-b}$$

Recall that our empirical postulate states that $b \in \{0, 1, 2\}$. Consequently, the denominator $(3-b)$ is strictly bounded and can only take the values 3, 2 or 1. For an integer solution d_1 to be generated, it is imperative that the numerator complies with an elementary divisibility condition dictated by the value of b:

- If $b = 2$, the denominator is 1: the value $d_1 = a \cdot r + d_2(q+1)$ is always a natural integer.
- If $b = 1$, the denominator is 2: the quantity $(a \cdot r + d_2(q+1))$ must strictly be even.

- If $b = 0$, the denominator is 3: the quantity $(a \cdot r + d_2(q+1))$ must strictly be a multiple of 3.

The algorithmic impact of this formulation is major. Since the law of asymmetry guarantees that the micro-divisor d_2 is extremely small compared to d_1, it is no longer necessary to factorize K nor to calculate the set of its divisors (an operation that is traditionally costly for very large numbers).

For a tested parameter c, the Euclidean division $n = qc + r$ immediately fixes the constants q and r. The algorithm then comes down to iterating over the first integer values of the micro-divisor d_2 (e.g., 1, 2, 3...) and the multiplier a. The determination of the macro-divisor d_1 is then performed by a direct evaluation calculation in constant time $\mathcal{O}(1)$.

It is ultimately sufficient *in fine* to check if the candidate d_1 thus generated actually divides K. This discovery transforms a heuristic search by factorization into a targeted and deterministic arithmetic generation.

6 The Ultimate Diophantine Reduction for $n \equiv 1 \pmod{24}$

It is obvious that elementary polynomial identities allow solving the conjecture for vast classes of congruences without algorithmic recourse. For example, if $n \equiv 3 \pmod 4$, the identity $\frac{4}{n} = \frac{1}{\alpha} + \frac{1}{\alpha n}$ with $\alpha = \frac{n+1}{4}$ provides an immediate decomposition. By the Chinese Remainder Theorem, the resistant class and unique hard core is $n \equiv 1 \pmod{24}$.

Lemma 5 : *For any prime n, let $g = GCD(d_1, d_2)$ and we have that $g = 1$*

Proof : Let $g = \mathrm{GCD}(d_1, d_2)$. Let us set $d_1 = g \cdot D_1$ and $d_2 = g \cdot D_2$, where D_1 and D_2 are coprime. Their LCM then becomes $g \cdot D_1 \cdot D_2$. By definition, K is a multiple of this LCM, therefore $K = m \cdot g \cdot D_1 \cdot D_2$ with $m \in \mathbb{N}^*$. Let us return to our initial equation $n = 4K - c$. To simplify the

initial analysis, let us place ourselves in the fundamental case where $a = 1$, which gives $c = d_1 + d_2$. By replacing K, d_1 and d_2 with their expressions:

$$n = 4(m \cdot g \cdot D_1 \cdot D_2) - gD_1 - gD_2 \iff n = 4mgD_1D_2 - gD_1 - gD_2$$

Let us look closely at the right side of the equation. All the terms contain g. Let us factor by g:

$$n = g \cdot (4mD_1D_2 - D_1 - D_2)$$

Since m, D_1 and D_2 are integers, the entire block within the parentheses is an integer (let's call it A). The equation simply becomes:

$$n = g \cdot A$$

The revelation is here: this equation literally means that g is an exact divisor of n. This is not an assumption, it is an algebraic obligation! Since we are studying cases where n is a prime number, the only possible divisors of n are 1 and n. Let us show by contradiction that $g = n$ is impossible. If $g = n$, that would mean that the GCD of d_1 and d_2 is n. Therefore d_1 and d_2 would both be multiples of n. Since we are taking distinct divisors, their sum $c = d_1 + d_2$ would therefore be at least $n + 2n = 3n$. Now we have proven that our search window is $c \leq 1.5K$, and since $K \approx 0.25n$, c is much smaller than n. It is physically impossible for c to be equal to $3n$. Implacable conclusion: Since g divides n, that n is prime, and that $g = n$ is too large, g is obligatorily equal to 1.

Remark : This is why the equation simplifies masterfully. Knowing that $g = 1$, let us return to $n = 4mD_1D_2 - D_1 - D_2$ and multiply everything by $4m$:

$$4mn = 16m^2D_1D_2 - 4mD_1 - 4mD_2$$

By adding 1 to both sides, we obtain the perfect final equation:

$$4mn+1 = 16m^2D_1D_2 - 4mD_1 - 4mD_2 + 1 \iff 4mn+1 = (4mD_1 - 1)(4mD_2 - 1)$$

Theorem 13 : *By applying the law of asymmetry and the modulator a, for any prime number $n \equiv 1 \pmod{24}$, there exists a pair $(a, m) \in (\mathbb{N}^*)^2$ such that $(4am - 1) \mid (an + 1)$.*

Proof : We have rigorously proven that $g = 1$. To respect our law of asymmetry ($d_2 \ll d_1$), let us place ourselves in the case where $D_2 = 1$ (micro-divisor) and $D_1 = D$ (macro-divisor). Let us reintegrate the multiplier a such that $a \cdot c = d_1 + d_2 = D + 1$. Thus $c = \frac{D+1}{a}$. According to our generating equation:

$$n = 4mD - \frac{D+1}{a} \iff an = 4amD - (D+1) \iff an = 4amD - D - 1$$

$$\iff an + 1 = 4amD - D \iff an + 1 = D(4am - 1)$$

For D to be a valid integer, it is imperative that the term $(4am - 1)$ exactly divides $(an + 1)$.

Theorem 14 : *For any quotient $Q \geq 1$ such that $an + 1 = Q(4am - 1)$, we have the absolute inequality $4m \leq n + Q$.*

Proof : The divisibility is algebraically equivalent to:

$$(4am - 1) \mid (an + 1) \iff (4am - 1) \mid (an + 1) + (4am - 1) - 4am + 4m$$

$$\iff (4am - 1) \mid (n + 4m)$$

Let us apply the quotient Q to this second formulation:

$$n + 4m = Q(4am - 1) \iff n + 4m = 4amQ - Q$$

$$\iff n + Q = 4amQ - 4m \iff n + Q = 4m(aQ - 1)$$

Since $a \geq 1$ and $Q \geq 1$, the term $(aQ - 1) \geq 1$. The equation therefore forces the absolute inequality $4m \leq n + Q$.

Lemma 6 : *The product of two integers $(4m)$ and $(aQ - 1)$ yielding $(n + Q)$ imposes that one of the two factors be less than or equal to the*

square root of their product.

Proof : Let the equation be $n + Q = 4m(aQ - 1)$. Let us set $x = 4m$ and $y = aQ - 1$. Let us assume by contradiction that $x > \sqrt{n+Q}$ and $y > \sqrt{n+Q}$. By multiplying these two strictly positive inequalities, we obtain:

$$x \cdot y > \sqrt{n+Q} \times \sqrt{n+Q} \iff x \cdot y > n + Q$$

Now we know that $x \cdot y = n + Q$, which leads us to a contradiction. Thus, one of the two factors is necessarily less than or equal to $\sqrt{n+Q}$.

By applying this property to the hard core, let us set $n = 24k + 1$ with $k \in \mathbb{N}^*$. The analysis leads us to distinguish two strict sub-cases.

6.1 Case 1: Bound on the multiplier $4m \leq \sqrt{n+Q}$

If the term $(4m)$ constitutes the smaller factor, the search for the parameter m is drastically restricted to an interval of the order of $\mathcal{O}(\sqrt{n})$.

Theorem 15 : *For $4m \leq \sqrt{n+Q}$, the validity of the solution requires that $(4am - 1) \mid (24k + 4m + 1)$.*

Proof : By algebraically isolating Q in our symmetric equation, the equation becomes:

$$n + 4m = Q(4am - 1) \iff Q = \frac{n + 4m}{4am - 1}$$

For Q to be an integer, the divisibility $(4am - 1) \mid (n + 4m)$ is required. By injecting the congruence of $n = 24k + 1$, we obtain:

$$(4am - 1) \mid (24k + 1 + 4m) \iff (4am - 1) \mid (24k + 4m + 1)$$

Although this case provides an excellent safety net, it proves insufficient to prove universality towards infinity.

6.2 Case 2: Bound on the quotient $aQ - 1 \leq \sqrt{n+Q}$

It is under the assumption that the term $(aQ-1)$ is the smaller factor that our reduction reveals its deterministic structure.

Theorem 16 : *By positing the change of variable $d = aQ - 1$, if $d \equiv -1 \pmod{a}$, then we fall exactly back onto the canonical form $an+1 = d(4am-1)$.*

Proof : By definition, d is a strictly positive integer. Let us isolate Q:

$$d = aQ - 1 \iff d + 1 = aQ \iff Q = \frac{d+1}{a}$$

For Q to remain an integer, it is imperative that a divides $(d+1)$, which translates to:

$$d + 1 \equiv 0 \pmod{a} \iff d \equiv -1 \pmod{a}$$

Let us reinject this expression for Q into our symmetric equation $n + Q = 4md$:

$$n + \frac{d+1}{a} = 4md \iff \frac{an+d+1}{a} = 4md \iff an + d + 1 = 4amd$$

$$\iff an + 1 = 4amd - d \iff an + 1 = d(4am - 1)$$

7 The Modulating Role of the Parameter a

Theorem 17 : *For any $k \geq 1$, the resolution requires the existence of a multiplier $a \geq 1$ such that the integer $a(24k+1)+1$ admits a decomposition into two factors $(d \times D)$ satisfying $D \equiv -1 \pmod{4a}$ and $d \equiv -1 \pmod{a}$.*

Proof : Let $n = 24k + 1$. According to our canonical form $an + 1 = d(4am - 1)$, let us set $D = 4am - 1$. We indeed have $an + 1 = d \times D$. Let

us verify the congruences of our two factors:

$$D = 4am - 1 \iff D + 1 = 4am \iff D \equiv 0 \pmod{4a} \iff D \equiv -1 \pmod{4a}$$

Furthermore, we saw during the change of variable that $d = aQ - 1 \iff d \equiv -1 \pmod{a}$.

The parameter a is therefore not a simple adjustment variable, but a true modulator of congruences. When a prime number n escapes the trivial case $a = 1$, incrementing a structurally modifies the factorization of $(an+1)$ until it forces the appearance of a divisor D belonging to the resolving congruence class modulo $4a$.